

Chern-Simons Gravity: Models and Observations

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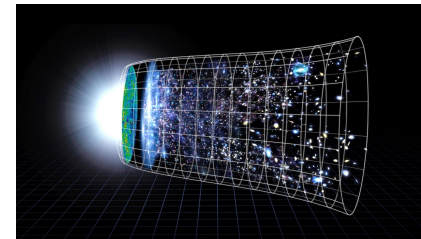
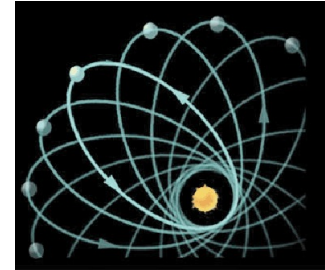
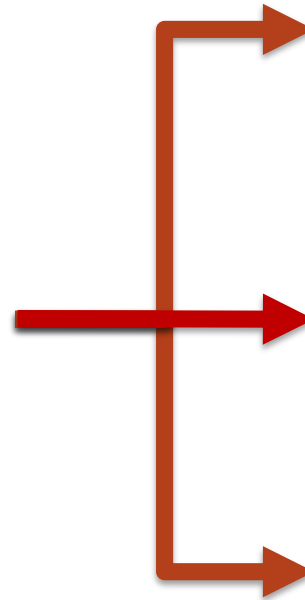
arXiv: 2207.05094, Collaborator: Stephon Alexander

General Relativity: The Einstein-Hilbert Action

$$S = \int d^4x \sqrt{-g} \kappa R$$

Spacetime
Metric,
Lorentzian
Signature

Ricci Scalar,
 $\kappa = (16\pi G)^{-1}$



What is (dynamical) Chern-Simons Gravity [dCS]?

$$S = \int d^4x \sqrt{-g} \left[\kappa R + \frac{a}{4f} {}^*R R - \frac{1}{2} (\nabla_\mu a) (\nabla^\mu a) \right]$$

↑

Spacetime
Metric,
Lorentzian
Signature

↑

Einstein-
Hilbert Term,
 $\kappa = (16\pi G)^{-1}$

↑

dCS Decay
Constant

↑

Pontryagin
Density
 ${}^*R R = {}^*R^\rho{}_\sigma{}^{\mu\nu} R^\sigma{}_{\rho\mu\nu}$
 ${}^*R^\rho{}_\sigma{}^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\alpha\beta} R^\rho{}_{\sigma\alpha\beta}$

↑

dCS (pseudo)scalar

dCS Model: Fermion Self-Interactions

What are Fermion Self-Interactions?

$$\mathcal{L}_\Psi = \bar{\Psi} (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi - \lambda \bar{\Psi} \Psi \bar{\Psi} \Psi$$

Diagram illustrating the components of the Fermion Lagrangian $\mathcal{L}_\Psi$:

- $i\gamma^\mu \partial_\mu$: Dirac Gamma Matrices
- $\tilde{m}_\Psi$: "bare" mass
- Ψ : Dirac Fermion
- λ : (Attractive) Fermion Self-Coupling Constant

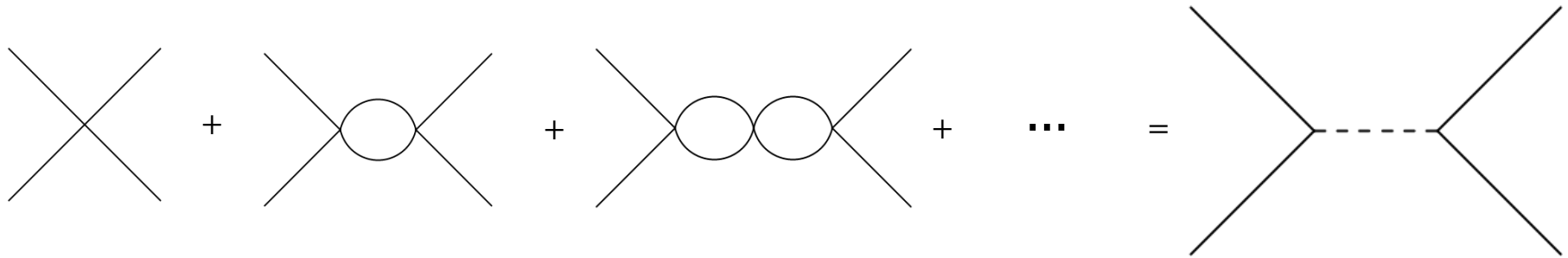
★ Cutoff Λ

How will we obtain dCS from Fermion-SI?

1. Identify and realize a scalar bound state in our theory of Fermion Self-Interactions (SI)
2. Demonstrate that the angular part of the (complex) scalar becomes the dCS scalar through fermion-loops with gravity after spontaneous symmetry breaking.

How is there a bound state?

Geometric Summation to identify pole



How to realize the bound state?

$$Z_\alpha = \int \mathcal{D}\alpha \mathcal{D}\bar{\alpha} \left(- \int d^4x \tilde{m}_\Phi^2 \bar{\alpha} \alpha \right)$$

$$\Psi = \Psi_\ell + \Psi_s$$

$$\Phi = \alpha - \tilde{m}_\Phi^{-2} \bar{\Psi}_s \Psi_s$$

$\Rightarrow$

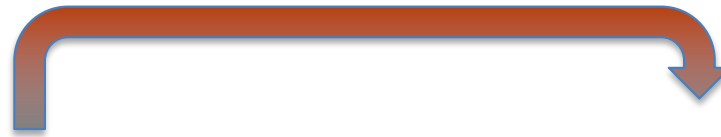
$$\begin{aligned} \tilde{\mathcal{L}}_\Psi = & \bar{\Psi} (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi - \lambda \bar{\Psi} \Psi \bar{\Psi} \Psi \\ & + (\partial_\mu \Phi^*) (\partial^\mu \Phi) - y_\Phi (\Phi \bar{\Psi} \Psi + \text{h.c.}) \\ & + m_\Phi^2 |\Phi|^2 - \frac{\lambda_\Phi}{4} |\Phi|^4, \end{aligned}$$

Integrate Out Short – Scale Modes

$$y_\Phi \sim \lambda \Lambda^2$$

Spontaneous Symmetry Breaking

$$\Phi = (1/\sqrt{2})(F + \sigma) \exp[ia(x)/F]$$



$$\mathcal{L}_\Phi = \bar{\Psi}_L (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi_L + \bar{\Psi}_R (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi_R \\ + \partial_\mu \Phi \partial^\mu \Phi^* - V(|\Phi|^2) - (y_\Phi \Phi \bar{\Psi}_L \Psi_R + \text{h.c.}),$$

$$\mathcal{L}_\Phi = \bar{\Psi} \left[i\gamma^\mu \partial_\mu - \left(\tilde{m}_\Psi + \frac{y_\Phi F}{\sqrt{2}} \right) \right] \Psi \\ + \frac{1}{2} \partial_\mu a \partial^\mu a - \frac{y_\Phi}{\sqrt{2}} a \bar{\Psi} \gamma^5 \Psi,$$



The Gap Equation



$$\frac{\Delta m_\Psi}{m_\Psi} \left(\frac{2\pi^2}{\lambda\Lambda^2} \right) = 1 - \frac{m_\Psi^2}{\Lambda^2} \log \left(1 + \frac{\Lambda^2}{m_\Psi^2} \right), \quad \Delta m_\Psi = y_\Phi F / \sqrt{2}$$

$\Lambda \gg m_\Psi$

$$F \approx 0.9m_\Psi$$

$\Lambda \ll m_\Psi$

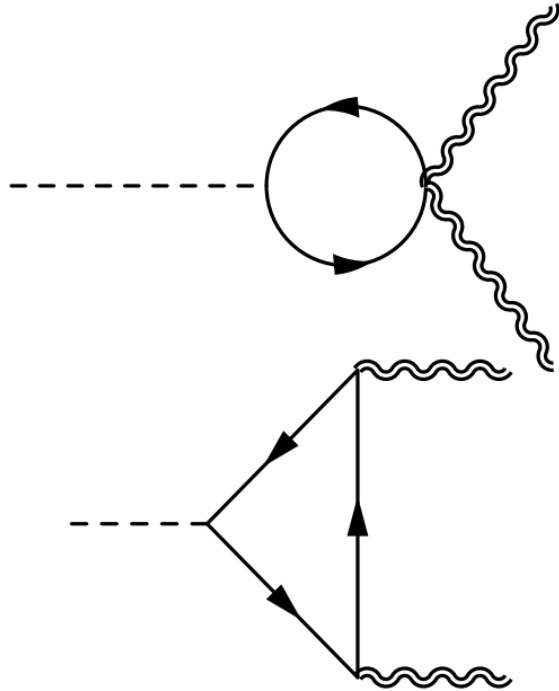
$$F \approx 0.45\Lambda^2/m_\Psi$$

Remember!



$$y_\Phi \sim \lambda\Lambda^2$$

Loop Generation of dCS

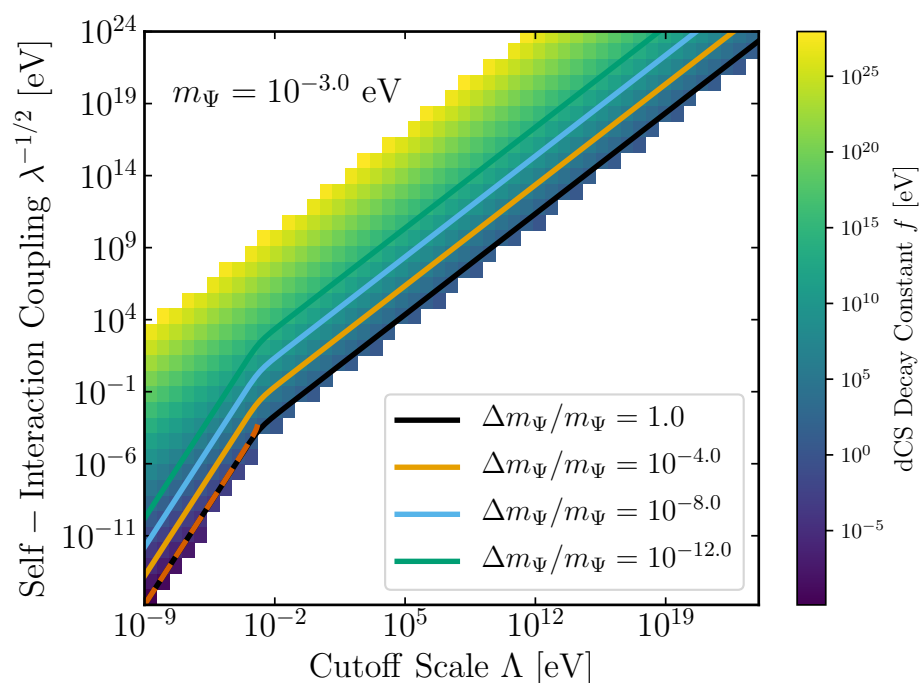


$$\mathcal{L}_g^{\text{eff}} = -\frac{g}{384\pi^2} \frac{a}{2m_\Psi} * RR$$

$$f = 192\pi^2 \frac{m_\Psi}{g}$$

$$g = y_\Phi/\sqrt{2}$$

General Parameter Space



$$\Lambda \gg m_\Psi$$

$$f = 1.7 \text{ eV} \left(\frac{\Delta m_\Psi}{m_\Psi} \right)^{-1} \left(\frac{m_\Psi}{10^{-3} \text{ eV}} \right)$$

$$\Lambda \ll m_\Psi$$

$$f = 0.85 \text{ eV} \left(\frac{\Delta m_\Psi}{m_\Psi} \right)^{-1} \left(\frac{\Lambda}{m_\Psi} \right) \left(\frac{\Lambda}{10^{-3} \text{ eV}} \right)$$

UV Completions for Fermion SI

(Scalar) Yukawa

$$\mathcal{L} \supset g_\chi \chi \bar{\Psi} \Psi$$

$$\Lambda \sim m_\chi$$

$$\lambda = (g_\chi / m_\chi)^2$$

$$f = 34 \text{ eV } g_\chi^{-2} \left(\frac{m_\Psi}{10^{-3} \text{ eV}} \right)$$

Gravitational Torsion

$$\mathcal{L} \supset (2/\gamma) R^{\mu\nu} \wedge e_\mu \wedge e_\nu$$

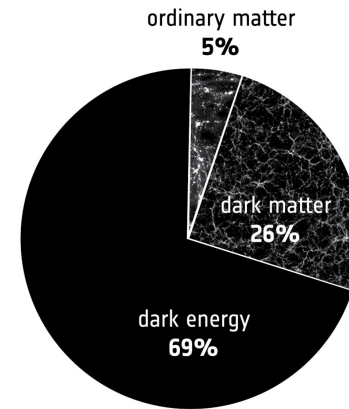
$$\Lambda \sim M_{\text{pl}}$$

$$\lambda = 3\pi / \Lambda_T^2$$

Fermion Candidates



Source: <https://stringfixer.com/files/26342399.jpg>



Source:
https://cdn.sci.esa.int/documents/33220/35293/Cosmic_energy_energy_budget.jpg/840136ea-a65b-125d-e421-82d2ae889c06?version=1.0&t=1572356777315

Summary

1. If fermions self-interact, a complex scalar bound state forms
2. The bound state undergoes spontaneous symmetry breaking, giving a Yukawa term
3. The Yukawa term generates dCS through loops interactions with gravity
4. Light fermions, such as neutrinos (and possibly DM), are ideal candidates

WIP

dCS Observations: Parity-Odd Galaxy Four-Point Function

What are Galaxy N-Point Functions (NPF)?

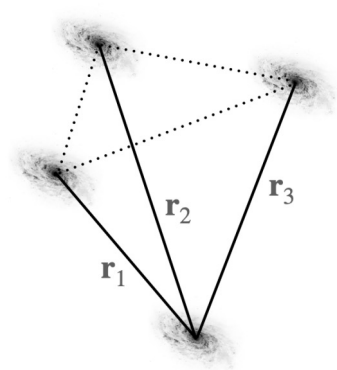
$$\delta_g(\mathbf{x}) = \frac{n_g(\mathbf{x}) - \bar{n}_g}{\bar{n}_g}$$

$$\langle \delta_g(\mathbf{x}) \rangle = 0$$

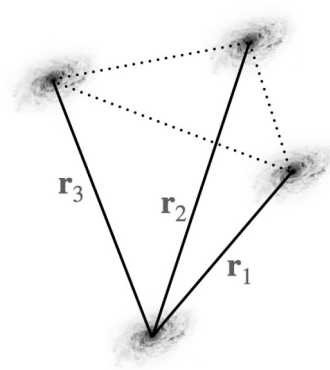
$$\langle \delta_g(\mathbf{x}_1) \delta_g(\mathbf{x}_2) \rangle \equiv \xi(\mathbf{x}_1, \mathbf{x}_2) = \xi(|\mathbf{x}_1 - \mathbf{x}_2|)$$

$$\langle \delta_g(\mathbf{x}_1) \dots \delta_g(\mathbf{x}_N) \rangle \equiv \xi(\mathbf{x}_1, \dots, \mathbf{x}_N)$$

Parity and the Galaxy 4PF



$$\zeta(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3)$$



$$\mathbb{P}[\zeta(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3)]$$

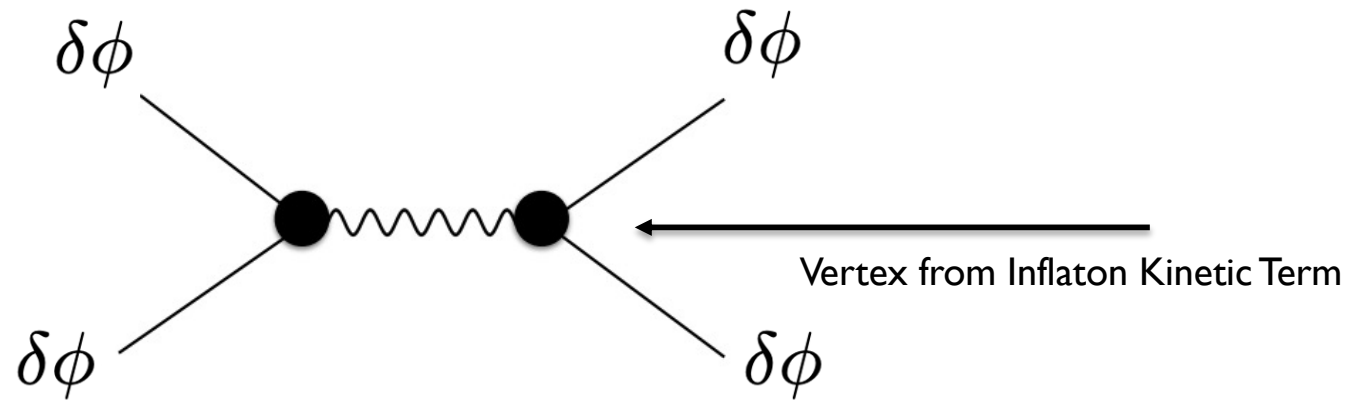
Inflationary dCS

$$S = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} (\partial_\mu \phi) (\partial_\nu \phi) + \kappa R + \frac{\phi}{4f} {}^*RR \right]$$

↑
Inflaton Kinetic Term

↑
Chooses Handedness for Gravitons
 $h_{\mu\nu} = (g_{\mu\nu} - \eta_{\mu\nu})/a^2$

Primordial Parity-Violating Trispectrum in dCS



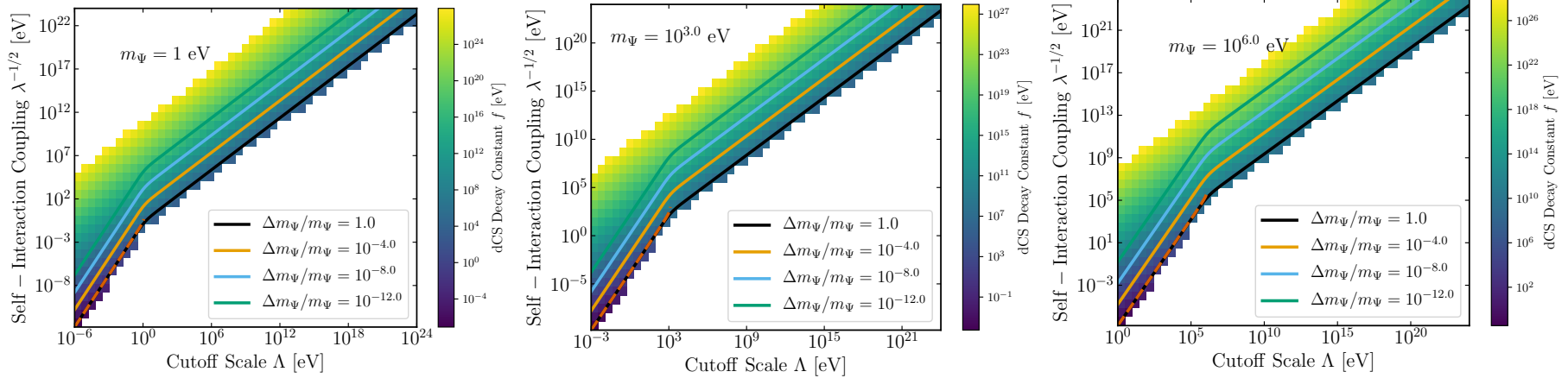
$$\langle \delta\phi(\mathbf{k}_1)\delta\phi(\mathbf{k}_2)\delta\phi(\mathbf{k}_3)\delta\phi(\mathbf{k}_4) \rangle_c^{\text{odd}} = \text{Im} \langle \delta\phi(\mathbf{k}_1)\delta\phi(\mathbf{k}_2)\delta\phi(\mathbf{k}_3)\delta\phi(\mathbf{k}_4) \rangle_c$$

Summary

1. In dCS inflation, one handedness of gravitons is chosen, yielding a parity-odd inflaton 4PF
2. Post-inflation, the inflaton decays into SM particles, turning off any parity-oddness in gravity
3. Inflation fluctuations (which are parity-odd) are inherited by SM particle density fluctuations
4. The resulting gravitational evolution of density fluctuations into galaxies is parity-even, preserving the primordial parity-odd signal

Backup Slides

More Parameter Space Plots



Exclusion Plots Neutrino Self-Interactions

